

AWS Auto Scaling Groups (ASG)

Why ASG: Automatically adjusts EC2 instances to match demand, ensuring availability and cost efficiency.

Why terminate old instances: To replace unhealthy, outdated, or non-compliant instances and maintain desired state.

Metrics used: CPU utilization, network traffic, request count, custom CloudWatch metrics.

Before ASG (On-Prem): Manual scaling, scripts, load balancers with static servers.

Scalability: Ability to handle increased load.

Availability: System uptime and accessibility.

High Performance: Fast response under load.

Resiliency: Ability to recover from failures.

Observability: Monitoring via logs, metrics, and traces.

Launch Template: Blueprint defining AMI, instance type, SGs; mandatory for ASG to launch instances.

Scale Out vs Scale In: Scale out adds instances; scale in removes instances.

Desired/Min/Max: Desired is target count, Min is lowest allowed, Max is highest allowed.

Health Checks: EC2 checks instance status; ELB checks application health via load balancer.

Termination traffic handling: ASG deregisters instance from ELB and waits for connections to drain.

Manual/Scheduled/Dynamic Scaling: Manual is user-driven, Scheduled is time-based, Dynamic is metric-based.

Cooldown period: Time between scaling actions to prevent rapid fluctuations.

Multi-AZ ASG: Distributes instances across AZs for high availability.

AWS VPC

VPC: Virtual private network in AWS to isolate and control networking.

Default vs Custom VPC: Default is auto-created with public subnets; Custom is fully user-defined.

CIDR block: IP range definition; plan to avoid overlap and allow growth.

Public vs Private IP: Public is internet-routable; Private is internal only.

Public vs Private Subnet: Public has IGW route; Private does not.

Route Table: Controls traffic routing within VPC.

Internet Gateway: Enables internet access for public subnets.

NAT Gateway vs Instance: Gateway is managed, scalable; Instance is manual and legacy.

Why NAT in public subnet: Needs IGW access to reach internet.

If NAT fails: Private instances lose outbound internet access.

Security Groups: Stateful firewall at instance level.

NACLs: Stateless firewall at subnet level.

Stateful vs Stateless: Stateful tracks connections; Stateless does not.

Traffic flow: Controlled by route tables, SGs, and NACLs.

VPC Peering: Connects VPCs privately; used for simple connectivity.

Peering limits: No transitive routing, CIDR overlap not allowed.

Transit Gateway: Central hub for connecting many VPCs.

TGW vs Peering: TGW supports transitive routing; peering does not.

VPC Endpoints: Private access to AWS services without internet.

Gateway vs Interface Endpoint: Gateway for S3/DynamoDB; Interface uses ENIs.

PrivateLink: Secure service exposure via interface endpoints.

DNS in VPC: Provided by AWS resolver via DHCP.

DNS hostnames vs resolution: Hostnames assign names; resolution enables DNS queries.

Elastic IP: Static public IP for stable access.

Change VPC CIDR: Add secondary CIDR block only.

On-prem connectivity: VPN or Direct Connect.

Site-to-Site vs Client VPN: Network-to-network vs user-to-network.

Direct Connect: Dedicated private connection to AWS.

High availability VPC: Multi-AZ subnets and redundant components.

Bastion Host: Secure jump server for private access.

Secure private instances: Use NAT, bastion, endpoints, SGs.

Flow Logs: Capture IP traffic for monitoring and troubleshooting.

Connectivity issues: Route tables, SGs, NACLs, IGW/NAT misconfig.

No internet access: Check route table, IGW, SG, NACL, NAT.

Restrict outbound access: NACLs, SG rules, no NAT route.

Source/Destination check: Disabled for NAT/Firewall instances.

Multi-AZ VPC: Subnets span AZs for resilience.

Subnet sizing: Ensures enough IPs for scaling.

IPv4 vs IPv6: IPv4 limited; IPv6 provides large address space.

IPv6 routing: Uses IGW only, no NAT needed.

VPC Sharing: Share subnets across AWS accounts.

VPC limits: CIDR size, subnets, routes per table.

Environment isolation: Separate VPCs or accounts.

Multi-tier VPC: Web, app, DB layers in separate subnets.

ALB vs NLB: ALB is layer 7; NLB is layer 4.

Secure VPC design: Least privilege, private subnets, endpoints.

Cross-region connectivity: Peering, TGW, VPN.

Internet to private EC2: Internet → ALB → Private EC2.

VPC Endpoint use cases: S3 access, private AWS APIs.

Production VPC: Multi-AZ, ALB, ASG, NAT, endpoints, monitoring.

Additional Topics

Subnet masking: Defines network and host bits.

Usable IPs: $2^{(32-\text{mask})}$ minus 5.

AWS reserves IPs: Network, router, DNS, future use.

/24 vs /26 vs /28: 251, 59, 11 usable IPs respectively.

Future scalability: Use larger CIDRs and multiple subnets.

DHCP in VPC: Assigns IP, DNS, gateway info.

DHCP options: DNS servers, domain name, NTP.

Modify DHCP: Yes, via new DHCP options set.

DNS depends on DHCP: Resolver info comes from DHCP.

CIDR planning: Avoid overlap, allow expansion.

Overlapping CIDR: Peering and routing fails.

Secondary CIDR: Added to extend IP range.

IPv4 vs IPv6 planning: IPv6 removes IP exhaustion.

TGW simplification: Centralized routing hub.

Hub-and-spoke routing: Traffic flows via central TGW.

TGW limits: Cost, route limits, complexity.

Flow Logs capture: Accepted, rejected traffic metadata.

Flow Logs storage: CloudWatch, S3.

Flow Log use cases: Security analysis, troubleshooting.

Avoid VPC peering: Large-scale or transitive needs.

No transitive routing: Peering is point-to-point only.

Cross-account peering: Share services across accounts.

Peering troubleshooting: CIDR overlap, routes, DNS, SGs.